0.1 Ideal

Definition 0.1.1. Let R be a Ring. A subset $I \subseteq R$ is called *ideal* of R if:

1. $I \subseteq R$ is a subgroup of R.
2. I is closed under the multiplication.
3. For any $r \in R$, $rI \subseteq I$ and $Ir \subseteq I$. (In other word, for any $r \in R, a \in I$, $ra \in I$ and $ar \in I$.)

Theorem 0.1.2. Let R be a Ring. Then, TFAE:

1. $I \subset R$ is an Ideal of R.
2. The additive Quotient Group $R/I \overset{\text{def}}{=} \{r + I \mid r \in R\}$ be a Ring under the operation:

$$(r + I) \times (s + I) = (rs) + I$$

Proof. Observation:

$$r_1 + I = r_2 + I \iff r_1 - r_2 \in I \iff \exists a \in I \ \text{ s.t } \ r_1 = r_2 + a$$

Now, for well-definedness, want to show that the equality

$$\begin{aligned} &(r + I) \times (s + I) = (rs) + I \\ \overset{(*)}{=} &[(r + \alpha) + I] \times [(s + \beta) + I] = (r + \alpha)(s + \beta) + I = (rs + r\beta + \alpha s + \alpha\beta) + I \end{aligned}$$

$(*)$ holds for any $r, s \in R$, $\alpha, \beta \in I$.
If I is Ideal, then $r\beta, \alpha s, \alpha\beta \in I$. Thus closed under the addition gives $(*)$.
Conversely, if this operation is well-defined, then for any $r, s \in R$, $\alpha, \beta \in I$, $(*)$ holds.
Substituting zero to each r, s, α, β gives I is ideal. $\square$

0.1.1 Properties of Ideal in Ring with identity

Definition 0.1.3. Let R be a Ring with identity, and $A \subseteq R$. Define *Ideal generated by* A as:

$$(A) \overset{\text{def}}{=} \bigcap_{\substack{I \text{ ideal} \\ A \subseteq I}} I$$

And,

$$RA \overset{\text{def}}{=} \{r_1a_1 + \cdots + r_na_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$
$$AR \overset{\text{def}}{=} \{a_1r_1 + \cdots + a_nr_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$
$$RAR \overset{\text{def}}{=} \{r_1a_1r_1' + \cdots + r_na_nr_n' \mid n \in \mathbb{N}, r_i, r_i' \in R, a_i \in A\}$$

Lemma 0.1.4. Let R be a Ring with identity, and $A \subseteq R$. Then, $(A) = RAR$.

Proof. Since RAR is ideal which contains A, $(A) \subseteq RAR$.
And, conversely, if $\sum_{i=1}^{n} r_ia_ir_i' \in RAR$, then $\sum_{i=1}^{n} r_ia_ir_i' \in (A)$ because each $r_ia_ir_i'$ are contained in (A), being (A) is ideal containing A and ideal is closed under the addition. □

Theorem 0.1.5. Let I be an ideal of Ring R with identity.

$I = R$ *if and only if* I contains a unit.

Proof. Right direction is clear by $1 \in R = I$.
Denote $u \in I$ be a unit with $vu = 1$, and Let $r \in R$ be given. Then,

$$r = r1 = rvu \in I$$

□

Definition 0.1.6. An Ideal M of R is *Maximal ideal* if: There is no Ideal I such that $M \subsetneq I \subsetneq R$.

Theorem 0.1.7. Let R be a Ring with identity.
Then, every proper ideal $I \subsetneq R$ is contained in a maximal ideal.

Proof. □

Lemma 0.1.8. Let R be a commutative Ring with identity, M, P are proper ideals of R.

1. M is Maximal Ideal *if and only if* R/M is a field.
2. P is Prime Ideal *if and only if* R/M is an integral domain.

Summary: M maximal $\iff$ R/M field $\implies$ R/M integral domain $\iff$ M prime.

Proof.

$$\begin{aligned}
M \text{ is maximal} &\iff \text{There is no ideal } I \text{ such that } M \subsetneq I \subsetneq R \\
&\iff \text{There are Ideals of } R/M \text{ only } 0 \text{ and } R/M \\
&\iff R/M \text{ is field}
\end{aligned}$$

$$\begin{aligned}
P \text{ is Prime Ideal} &\iff \text{If } ab \in P\text{, then } a \in P \text{ or } b \in P \\
&\iff \text{If } ab + P = P\text{, then } a + P = P \text{ or } b + P = P \\
&\iff \text{If } \overline{a}\overline{b} = \overline{0}\text{, then } \overline{a} = \overline{0} \text{ or } \overline{b} = \overline{0} \\
&\iff R/P \text{ is integral domain}
\end{aligned}$$

□